ZARAGOZA. LOGISTICS CENTER

MZ 505
Inventory Systems

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Session 16: The (s, S) Periodic-Review Inventory Policy

Some parts adapted from D. Simchi Levi et al., “The Logic of Logistics” (1997)

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What if we relax some assumptions of OUT model?

Demand / Customer

- Stationary demand
- Independent across periods
- Known parameters
- If stocked out, backorder

Supplier

- Lead time: deterministic and integer multiplier of review period
- Unlimited capacity
- Perfect quality

Costs & Environment

- Costs are known
- No set-up costs
- Long shelf-life
- No obsolescence

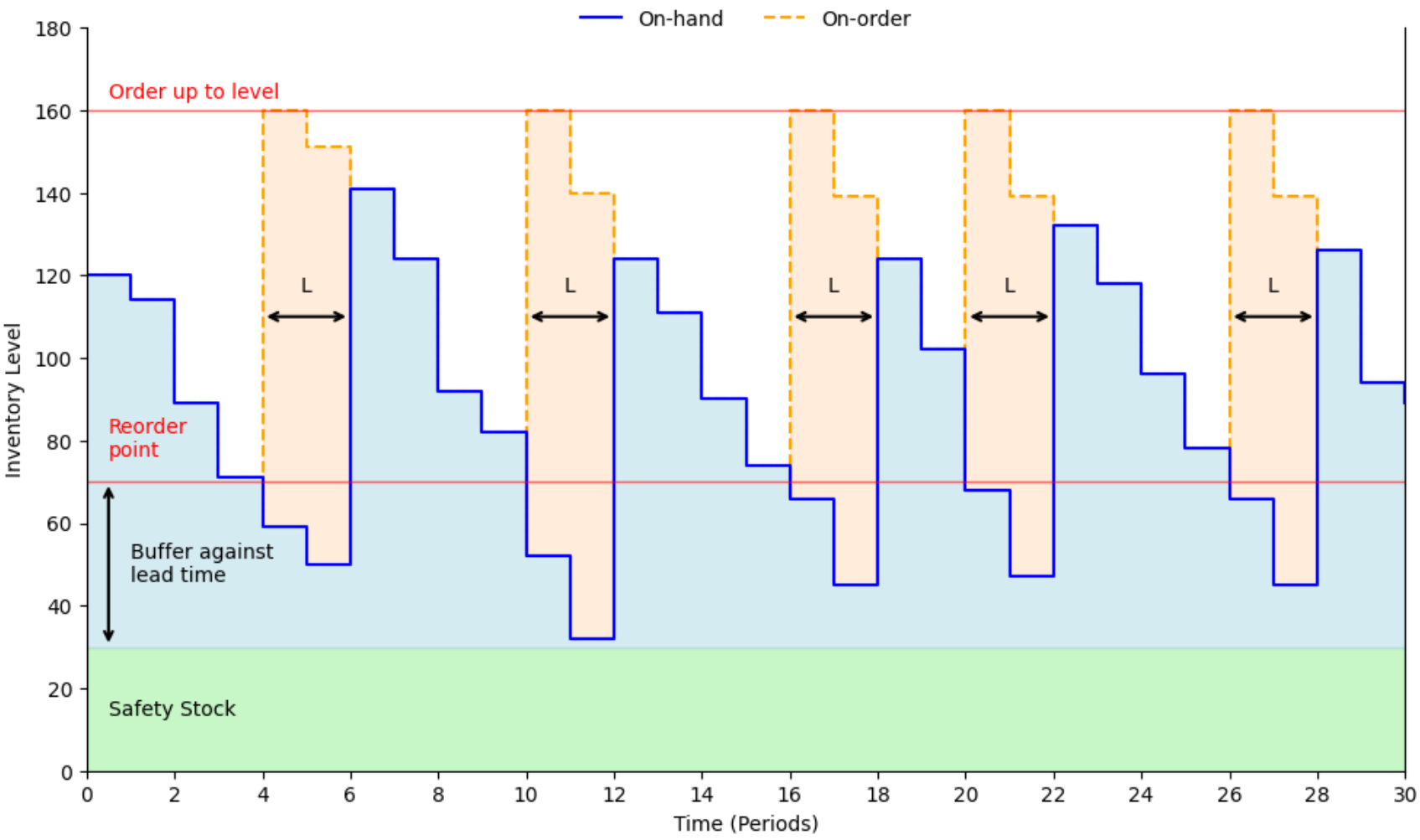
Inventory Policy

- Constant order-up-to point
- Constant review period
- Many replenishments over a long planning horizon

What if there is a set-up cost associated with each replenishment of inventory?

- … then there is a tradeoff between carrying little inventory (favored by short period lengths) and reducing ordering costs (favored by long period lengths).
- Increasing the period length leads to larger and less frequent orders:
  - Average order quantity = E[demand in a single period]
  - Frequency of orders ≈ 1 / (length of review period)
- The "optimal" review period is the one that minimizes the sum of ordering and holding cost per unit of time (e.g., per year)
  - See solution to HW3 Quiz Problem

The (s, S) Policy



Some motivation from the Newsvendor model

- Objective function: choose y to maximize the total **expected profit**

$$\pi(y) = E[p \cdot \min(y, D) + u \cdot (y - D)^+ - g \cdot (D - y)^+ - c \cdot y]$$

sales | salvaged | short | purchasing

- Then, the optimal y (let's denote it S) is given by: $F(S) = \frac{C_u}{C_u + C_o}$
- Note that, implicitly, three assumptions have been made in the above formulation:
  - There is no initial inventory
  - There is no fixed set-up cost for starting production
  - The excess demand is lost
- What if we relax the first two assumptions?

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Some motivation from the Newsvendor model (cont.)

- Assume there is some initial inventory, y_0, and a set-up cost, K.
- Then, the expected profit of producing $(S - y_0)$ units is

$$\pi(S) - K + cy_0$$

- Consider two cases:
  - If $y_0 \geq S$, we should not produce anything.
  - If $y_0 < S$, then it is optimal to bring inventory up to S only if

$$\underbrace{\pi(y_0) + cy_0}_{\text{Profit associated with not producing anything}} < \underbrace{\pi(S) - K + cy_0}_{\text{Profit associated with producing } S - y_0}$$

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The (s, S) Policy

- Let s be a number such that

$$\pi(s) = \pi(S) - K$$

- Then, the optimal policy in the (relaxed) Newsvendor problem has the following structure:
  - *If* $y_0 < s$, *order* $S - y_0$
  - *Otherwise, do not order*
- In multi-period environment, when the holding and penalty costs are constant, demand is stationary (infinite horizon), and there is no lead time in the delivery of orders (as in the online exercise we just played), then, an (s, S) inventory policy is optimal (Scarf, 1960)
  - …however, determining the optimal values of s and S is not so simple.

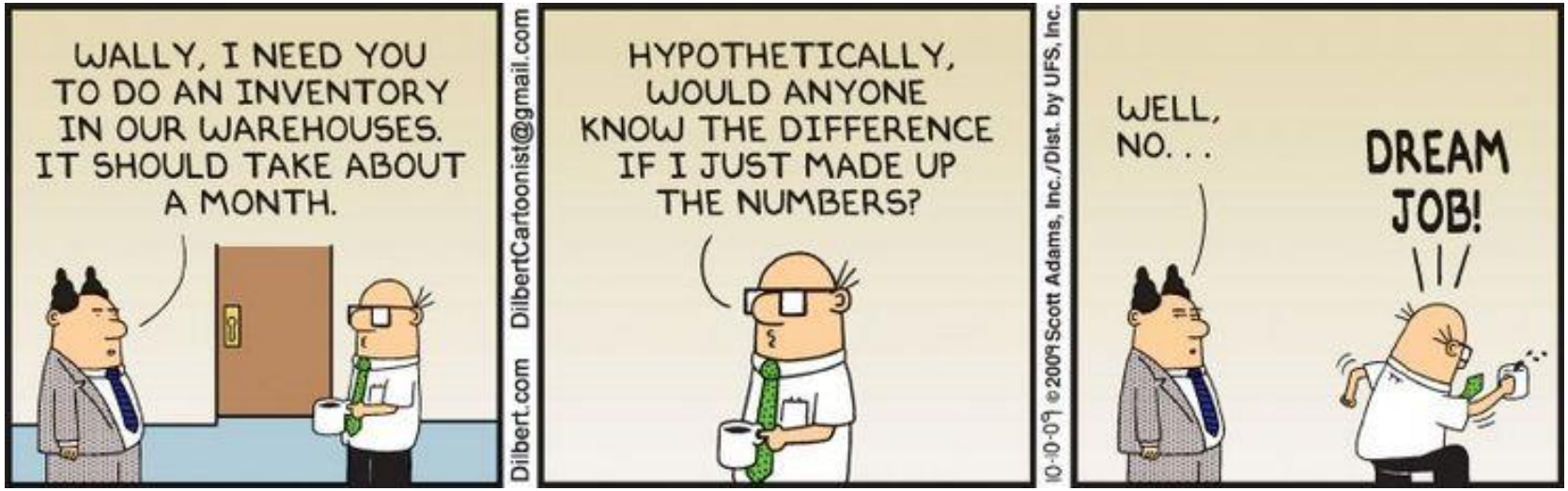
*The (s, S) Policy: A heuristic approach**

- First, let's try to find a good approximation for s:
  - Assume you are a Newsvendor facing only one period ahead.
  - Then, your optimal quantity is determined by the critical ratio (where the underage and overage values depend on your holding and backorder costs).
  - Set s such that $F(s) = \frac{C_u}{C_u + C_o}$ for <u>one-period</u> demand
- To approximate S:
  - Compute the EOQ and calculate the average number of review periods of coverage in EOQ.
  - Set S according to the order-up-to formula, $F(S) = \frac{C_u}{C_u + C_o}$ for demand corresponding to the <u>number of periods covered in EOQ</u>

* Relevant when EOQ covers demand for several review periods

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Coming up next: The (R, Q) Model



Thank you!